

A top-down view of a grey desk with various items: a silver stethoscope, a laptop displaying a medical website, a pair of glasses, a black smartphone, a white coffee cup, and a spiral-bound notebook with a black pen. A large, semi-transparent white pill shape is centered over the desk.

Pharm*Achieve*

RHEUMATOID ARTHRITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ABAT	Abatacept	HIV	Human Immunodeficiency Virus	Q	Every
ACE	Angiotensin Converting Enzyme	HLA	Human Leukocyte Antigen	RA	Rheumatoid Arthritis
AE	Adverse Effects	HTN	Hypertension	RTX	Rituximab
ASA	Aspirin	HZ	Herpes Zoster	Rxn	Reaction
AST/ALT	Aspartate/Alanine Transaminase	IL	Interleukin	SC	Subcutaneous
AZA	Azathioprine	IM	Intramuscular	Scr	Serum Creatinine
Bdmard	Biologic DMARD	IV	Intravenous	SJS	Steven Johnson Syndrome
CD	Cluster of differentiation	LEF	Leflunomide	SLE	Systemic Lupus Erythematosus
C/I	Contraindicated	LFT	Liver Function Test	SMP-TMX	Sulfamethoxazole-trimethoprim
CBC	Complete Blood Count	MHC	Major Histocompatibility Complex	SSRI	Selective Serotonin Reuptake Inhibitor
CRP	C-reactive Protein	MOA	Mechanism Of Action	SSZ	Sulfasalazine
Csdmard	Conventional Synthetic DMARD	N/V/D	Nausea/Vomiting/Diarrhea	TB	Tuberculosis
CV	Cardiovascular	MRI	Magnetic Resonance Imaging	TCZ	Tocilizumab
CXR	Chest X-ray	MSM	Men who have Sex With Men	Tsdmard	Targeted Synthetic DMARD
CYP	Cytochrome P	MTX	Methotrexate	URTI	Upper Respiratory Tract Infection
D/I	Drug Interactions	NSAID	Non-steroidal Anti-inflammatory Drug	WNL	Within Normal Limits
DMARD	Disease Modifying Antirheumatic Drug	OA	Osteoarthritis		
ESR	Erythrocyte Sedimentation Rate	OT/PT	Occupational/Physical Therapy		
G6PD	Glucose-6-phosphate Dehydrogenase	PFT	Pulmonary Function Tests		
GI	Gastrointestinal	PML	Progressive Multifocal Leukoencephalopathy		
HBV/HCV	Hepatitis B/C Virus	PO	By Mouth		
HCQ	Hydroxychloroquine	Pts	Patients		

DEFINITION & RISK FACTORS

DEFINITION

Rheumatoid Arthritis: Chronic, systemic, inflammatory, autoimmune disorder (destruction of synovial joints, tendons, bones, organs)

- **Symmetric & Polyarthritis** (multiple joints)



Approximately 1% of adults are affected

RISK/PROGNOSTIC FACTORS

Gender & Age

- **Females** higher risk (3:1)
- Can occur at any age; peak ages **30 and 50**

Family History and Genetics

- MHC/HLA genes play a role in RA

Ethnicity

- Traditionally thought to be most prevalent amongst Caucasians – now unclear if ethnicity is a risk factor

Environmental

- Smoking and sedentary lifestyle are poor prognostic factors

PATHOPHYSIOLOGY

FYI

Genetic Events

- Human leukocyte antigen
- Major histocompatibility complex

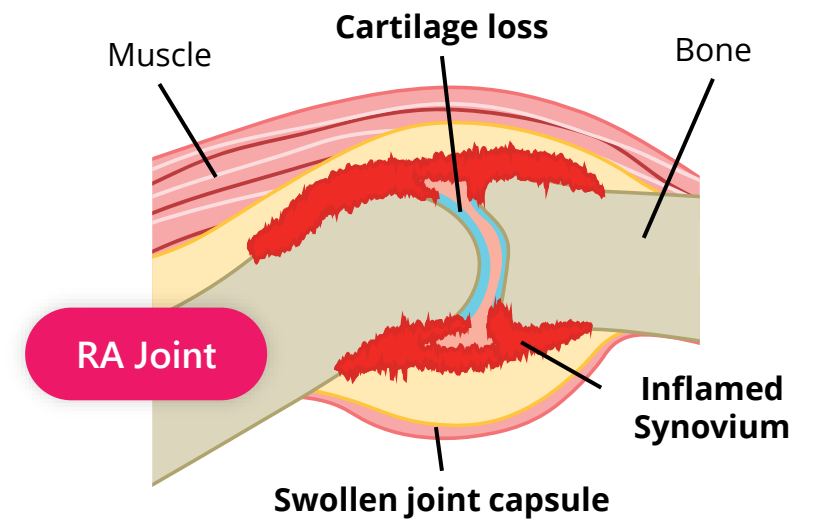
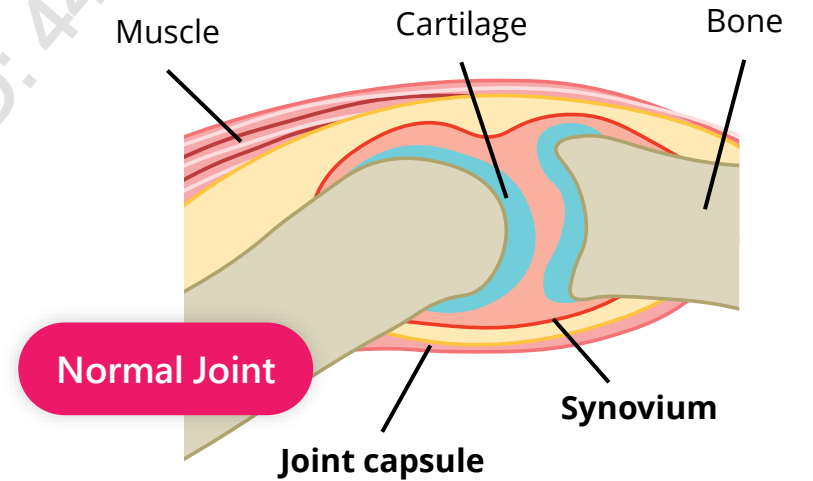


Stochastic Events

- Random environmental events
- Repeated activation of innate immunity

Production of **anti-citrullinated protein antibodies** that build up over time and repeated activation of immune system

Autoimmune process established, synovium organizes into invasive tissue that **degrades cartilage and bone**



CLINICAL PRESENTATION

Classic Symptoms

- Bilateral, symmetric
- Joint pain
- Morning stiffness **lasting longer than 30-60 minutes**
- Tenderness and swelling
- Redness and warmth
- Swan neck deformity in hands

Other Symptoms

- Fatigue
- Malaise
- Low-grade fever
- Loss of appetite
- Stiffness after staying in one position for too long
- **Improves with movement**

Complications

- Joint destruction
- Extra-articular manifestations:
 - Lymphoproliferative disease
 - Felty's syndrome (i.e. RA, splenomegaly and low WBC)
 - Pulmonary complications (e.g. pulmonary fibrosis)
 - **Sjogren's syndrome** (causes dry eyes and mouth)
 - Increased risk of **cardiovascular (CV) disease** and CV mortality
- Anemia
- Increased body fat
- Osteopenia
- Muscle weakness
- Vasculitis
- Rheumatoid nodules (firm lumps under the skin, can also form in the lungs)

JOINT INVOLVEMENT



Osteoarthritis



Rheumatoid arthritis

Distal phalanges

Proximal phalanges

Intermediate phalanges

Metacarpals

Carpals

Distal AND
Proximal interphalangeal
joints are mostly affected

Proximal AND
Metacarpophalangeal
joints are mostly affected

DIAGNOSIS

There is no one specific test to diagnose RA

History: severity of pain, duration of morning stiffness, limitation of function

Physical exam: number of swollen joints, extra-articular manifestations (e.g. dry eyes, rheumatoid nodules)

Investigations: radiographs of hands/feet, CRP, ESR, RF, anti-CCP (also known as anti-citrullinated protein antibodies)

- Symptoms present for **at least 6 weeks**

GOALS OF THERAPY

- ✓ Reduce or eliminate signs and symptoms
 - ✓ Pain, stiffness, fatigue
- ✓ Prevent/limit disease progression
 - ✓ No radiographic progression or evidence of further joint damage
- ✓ Maintain function and maximize quality of life
- ✓ Achieve remission or low disease activity
 - ✓ Target 50% clinical improvement within 3 months
 - ✓ Remission is ideal, but if not possible, target low disease activity within 6 months

NON-PHARMACOLOGICAL MEASURES



Exercise



Assistive devices: canes, grab bars, shower seats, joint splints: reduce inflammation and relieve symptoms



Weight loss: reduces stress on the joints



CV risk factor modification: smoking cessation, dyslipidemia, etc.

OT/PT/DIETITIAN/HOSPITAL REFERRAL
Evaluate daily living activities and function

- Nutrition/dietary therapy
- Surgery for advanced, severe RA unresponsive to treatment
- Physical therapy
 - After inflammation begins to subside
 - Specific exercises for joints to restore muscle mass and preserve strength

TREATMENT OVERVIEW

Primary goal: achieve remission OR, if not feasible, minimal disease activity

Treatment options: disease-modifying antirheumatic drugs (DMARD)

- csDMARDS (traditional or conventional synthetic), bDMARDS (Biologic), tsDMARDS (targeted synthetic)

Bridging therapy: used for early management of RA or bridging therapy

- Short-term use of glucocorticoids; lowest effective dose until DMARDS take effect

DMARDs (csDMARDs, bDMARDs, tsDMARDs)

- Initiate **csDMARD** as soon as possible
- **Slows disease progression** and reduces symptoms

Monitoring:

- Active disease: every **1-3 months**
- In remission: every **6-12 months**
- Adjust therapy after **3-6 months** if inadequate response
- Combination therapy may be considered

Adjunctive treatment (analgesics, glucocorticoids)

- NSAIDs do not alter disease progression and are intended as adjunctive therapy – **limit to a short duration**
- Opioids and acetaminophen lack evidence, but acetaminophen may be used due to its safety profile
- Steroids may be used for flare-ups, acute management and symptom control if necessary
 - Prednisone is effective, but many AEs

TREATMENT RECOMMENDATIONS

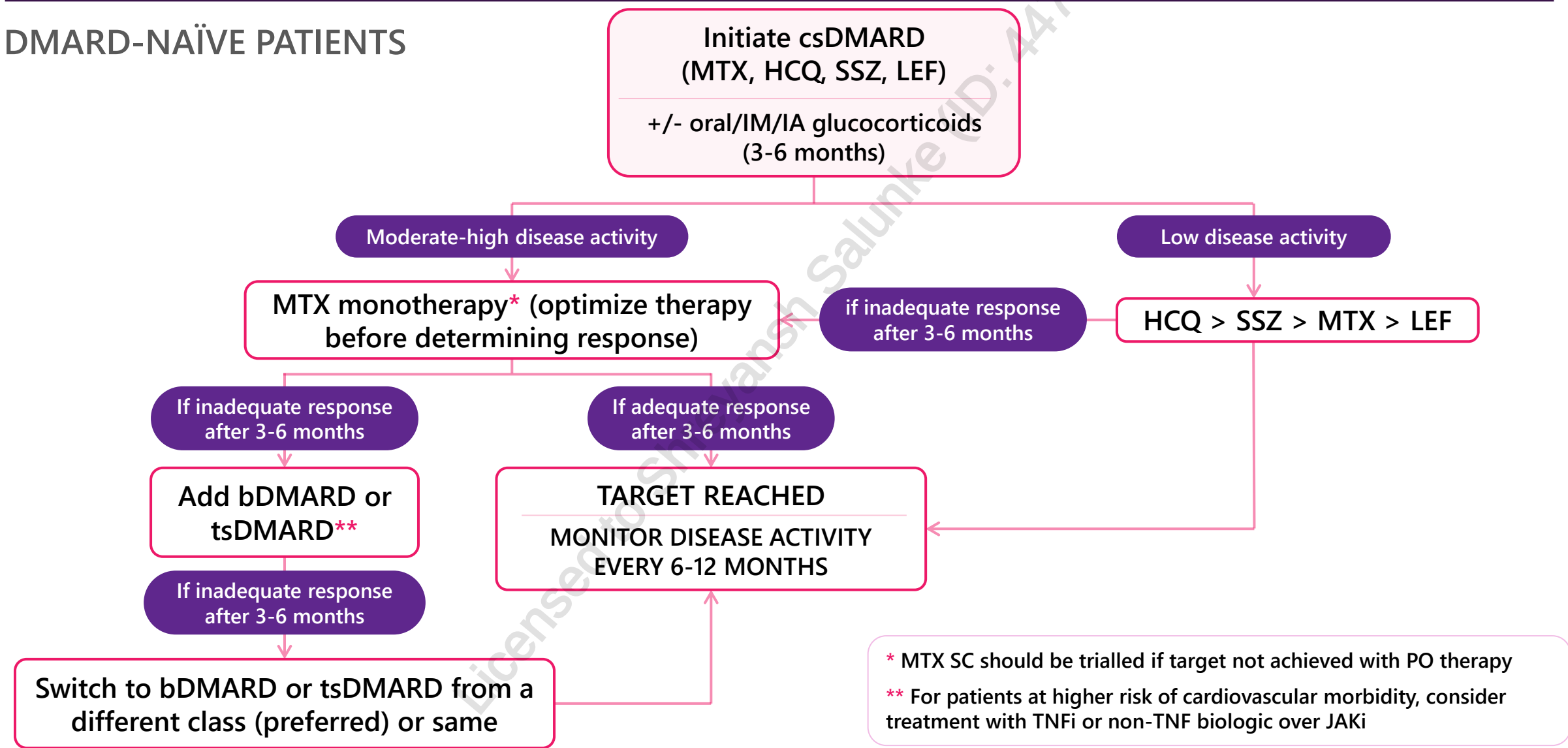
DMARD-NAÏVE PATIENTS

- Initiate csDMARD (MTX, HCQ, SSZ, LEF) as soon as possible tailored to disease activity and prognostic features
- May consider adjunctive glucocorticoids (oral, IM, IA) for < 3 months (strong recommendation)
 - Necessary to alleviate symptoms prior to onset of action of DMARDs
 - Should be limited to lowest effective dose for shortest duration possible

Disease Activity	Recommendation(s) as per 2021 ACR guidelines
Low	HCQ is conditionally recommended over other csDMARDs (SSZ > MTX > LEF)
Moderate-to-High	MTX monotherapy is strongly recommended over: - HCQ or SSZ - bDMARD or tsDMARD monotherapy - Combination of MTX + non-TNF bDMARD or tsDMARD MTX monotherapy is conditionally recommended over: - LEF - Dual or triple csDMARD therapy - MTX + TNFi

TREATMENT ALGORITHM

DMARD-NAÏVE PATIENTS



NSAIDS

DRUG	MOA	SAFETY	C/I
NSAIDS (naproxen, ibuprofen, diclofenac, celecoxib)	<u>Inhibits COX1:</u> anti-platelet effects <u>Inhibits COX2:</u> analgesic, anti-inflammatory, and antipyretic effects	AEs: • GI upset (N/V/D) • Gastric ulceration/bleeding • Renal dysfunction • Cardiovascular: chest pain, edema, HTN D/I: • Antihypertensives (NSAIDs may ↓ effect) • SSRI/SNRI (↑ risk of bleeding) • Corticosteroids (↑ risk of GI effects) • Antiplatelets, anticoagulants • Digoxin, lithium (may ↑ serum conc)	• Active peptic ulcer or GI bleed • Severe hepatic or renal impairment • Severe, uncontrolled HF • ASA: < 18 years with or recovering from viral illness (risk of Reye's Syndrome) • 3rd trimester of pregnancy (avoid in pregnancy if possible, especially third trimester)



CLINICAL PEARLS



AVOID/use NSAIDs cautiously in elderly, cardiac, renal and GI issues

- Used **for mild symptoms** as an analgesic and anti-inflammatory agent for **short-term**
- **Does NOT alter disease progression nor have any use in erosive disease**
- Gradually ↑ dose over first 1–2 weeks. If there is inadequate response or poor tolerance after 4 weeks, switch to a different NSAID

GLUCOCORTICOIDS

DRUG	MOA	AEs	D/I
Examples: prednisone , methylprednisolone acetate, triamcinolone	Potent anti-inflammatory that ↓ cytokines	• Insomnia • GI effects • Fluid retention, hypertension, dyslipidemia • Weight gain • Hyperglycemia • Osteoporosis • Apathy, depression • ↑ risk of infection after prolonged use	↑ risk of bleeding when combined with NSAIDs, antiplatelets, anticoagulants



CLINICAL PEARLS

BENEFITS

- May be used for acute symptom control during initiation of DMARD therapy & flare-up for active RA
- Works quickly within **1-3 days**

LIMITATIONS

- Many adverse effects
- Limited to short-term use

METHOTREXATE– 1ST LINE (MODERATE-SEVERE)

csDMARD

DRUG	MOA	AEs	D/I	Dose and administration
METHOTREXATE	Dihydrofolate reductase inhibitor	- N/V/D Dermatologic reactions - Headache, dizziness, fatigue - Oral ulcer - Cough Infection - Interstitial pneumonitis ↑ liver enzymes - Bone marrow suppression	- Leflunomide (must monitor for hepatotoxicity) - High-dose ASA (ASA 81 mg not of concern) Sulfonamides (if using, monitor CBC, LFTs and SCr) - Tacrolimus, pimecrolimus (enhances immunosuppressive effect of MTX) Alcohol may ↑ risk of hepatotoxicity - C/I: pregnancy, breastfeeding, severe renal impairment, chronic liver disease, pre-existing blood disorder	- PO or SC - Once weekly administration Titrate to at least 15 mg weekly within 4-6 weeks - PO doses may be split during the same 24-hour period to improve absorption and tolerability (doses > 15mg are not well absorbed) - Switching from PO to SC may improve tolerability



CLINICAL PEARLS

Supplement folic acid (at least 5 mg weekly) → to minimize side effects, including myelosuppression (usually given the day after methotrexate administration)

- Optimize MTX before adding or switching to other DMARDs**
- If targets are not reached with PO → switch to SC

BENEFITS

- “Gold standard” csDMARD – 1st line** for patients with moderate-to-high disease activity
- Effective as monotherapy or in combination with other DMARDs

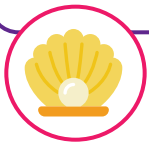
LIMITATIONS

- Slow onset of action → as early as **3-6 weeks**, up to 12 weeks
- Extensive adverse effect profile

SULFASALAZINE

csDMARD

DRUG	MOA	AEs	D/I
SULFASALAZINE	↓ synthesis of inflammatory mediators	• N/V/D, headache (use enteric-coated tablets to aid with GI effects) • Skin rash, sun sensitivity • Skin/urine discoloration (orange/yellow) • May cause oligospermia (no data suggesting it is teratogenic) • DRESS reaction (occurs within 3 months and usually resolves upon discontinuation) • Immunosuppression (less than MTX)	• Warfarin – changes in anticoagulant effect • NSAIDs – may enhance nephrotoxic effect • G6PD deficiency – use with caution due to risk of hemolytic anemia • May decrease serum conc of digoxin • C/I: sulfonamide or salicylate allergies, severe renal or hepatic impairment



CLINICAL PEARLS

- Onset of action is at least 4 weeks

BENEFITS

- Use for early, mild RA with low disease activity, considered a low potency DMARD
- Can be used instead of MTX for patients with C/I to MTX

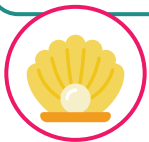
LIMITATIONS

- Less evidence than MTX for use as monotherapy
- Monitor LFTs, signs of infection, dermatological toxicity

HYDROXYCHLOROQUINE- 1ST LINE (MILD)

csDMARD

DRUG	MOA	AEs	D/I
HYDROXYCHLOROQUINE	DNA/RNA synthesis interference	• QTc prolongation with long-term use or high doses • Alopecia, abdominal pain, headache, dizziness, myopathy • Retinopathy with long-term use (serum conc-dependent) – eye exam at baseline and annually after 5+ years if no other risk factors • <u>Rare</u>: severe hypoglycemia, hemolysis (caution with G6PD deficiency)	• Antidiabetic agents – increased risk for hypoglycemia • Cimetidine • Cyclosporine • Metoprolol • Mefloquine • Quinolones • QTc prolonging agents • C/I: preexisting retinopathy



CLINICAL PEARLS

BENEFITS

- Use for initial therapy for **mild RA** with low disease activity
- Least toxic of all DMARDs due to lack of myelosuppressive effects, renal and hepatic toxicity

LIMITATIONS

- May require **several weeks** to respond; if no efficacy seen after 6 months, then consider therapy failure

LEFLUNOMIDE

csDMARD

DRUG	MOA	AEs	D/I
LEFLUNOMIDE	Inhibits pyrimidine synthesis acting on lymphocytes	• ↑ LFTs, severe liver injury • SJS/TEN, DRESS (monitor for skin/mucosal reactions) • Hematologic toxicity • Immunosuppression (↑ risk of infections) • Nausea, diarrhea • Anorexia, weight loss • Pulmonary fibrosis, interstitial lung disease • Headache • HTN • Peripheral neuropathy • Alopecia	• Other hepatotoxic drugs – e.g. alcohol (avoid), methotrexate • Other immunosuppressive drugs • Other hematotoxic drugs • Drugs metabolized by CYP1A2 – e.g. duloxetine • Drugs metabolized by CYP2C9 – e.g. phenytoin, warfarin • Combination therapy with MTX not recommended – if used together: leflunomide 20 mg po daily without loading dose and monitor continuously for hepatic toxicity (monthly LFTs) C/I: severe renal impairment, impaired liver function, breastfeeding, pregnancy



CLINICAL PEARLS

- Screen for TB prior to initiation

BENEFITS

- ↓ progression of erosions in RA; onset 4-6 weeks
- For patients who fail conventional treatment
- Can be used instead of MTX for patients with C/I to MTX

LIMITATIONS

- Long half-life - washout with cholestyramine or activated charcoal if rapid reduction in drug levels is required (e.g. cholestyramine 8g TID or activated charcoal 50g QID for 11 days)
- Teratogenic → R/O pregnancy prior to initiation, reliable contraception required for up to 2 years following D/C without washout

DMARD MONITORING

DMARD	MONITORING PARAMETERS
MTX	CBC, LFTs, SCr: baseline, then monthly for 1 st 3 months, then q1-3 months thereafter CXR, hepatitis B&C serology: baseline
LEF	CBC, LFTs*, SCr: baseline, then monthly for 1 st 3 months, then q1-3 months thereafter Hepatitis B&C serology: baseline *LFTs should be completed monthly if also taking MTX
SSZ	CBC, LFTs: baseline, then q3 months SCr: baseline
HCQ	Eye exams: baseline, then annually after 5 years of treatment (or annually from baseline if patient is high-risk*) *High-risk is defined as high doses (> 6.5mg/kg or > 400mg/day), cumulative doses > 1000 g, >5-7 years of treatment, older age, liver/kidney disease, obesity, or pre-existing eye conditions

TNF- α INHIBITORS

bdMARD

DRUG	MOA	AEs	D/I
ANTI-TNF α (infliximab, etanercept, adalimumab, golimumab, certolizumab pegol)	Blockade of TNF α	• <u>Common</u>: GI, acute injection site reactions, acute/delayed infusion reactions, headache, cough • Infection risk (TB, bacterial, viral, fungal) • ↑ risk of malignancies (i.e. lymphoma)	• Immunosuppressants • Tacrolimus, pimecrolimus • C/I: Active infection, moderate-severe HF – NYHA III or IV (use with caution in mild HF)



CLINICAL PEARLS

- Administered SC (except infliximab – IV only)

BENEFITS

- Used (alone or in combination w/ MTX) for pts **with moderate-severe** RA
- **Etanercept is approved for juvenile idiopathic arthritis**
- More effective in preventing radiographic damage when combined with MTX
- Varying onset depending on drug and individual but rapid improvement in lab parameters of inflammation usually within 8-12 weeks

LIMITATIONS

- \$\$\$
- Association with serious bacterial and opportunistic infections

MONITORING

- Baseline CBC, screen for TB, Hep B/C, signs/symptoms of infection

JAK INHIBITORS

tsDMARD

DRUG	MOA	AEs	D/I
Tofacitinib, Upadacitinib, Baricitinib (targeted synthetic DMARD)	JAK kinase inhibitor - block signals through several cytokines in inflammatory pathway	• Nasopharyngitis • Hyperlipidemia • Infections • HTN, headaches • N/D • Skin rash • Increased LFTs • Increased risk of malignancy (e.g. lymphoma, lung cancer)	• Potent immunosuppressants (e.g. azathioprine, tacrolimus, cyclosporine) • Potent CYP3A4 inhibitors • C/I: severe hepatic impairment, pregnancy, breastfeeding



CLINICAL PEARLS

- Administered orally
- Recommend HZ vaccination

LIMITATIONS

- FDA concluded **increased risk of major adverse CV events, thrombosis, cancer**
- Use with caution in patients at increased risk of **GI perforation**



Used (alone or in combination w/ MTX) for patients **with moderate to severe** RA if bDMARDS failed

MONITORING

- Signs of infection, HR, BP, abdominal symptoms
- **Screen for tuberculosis at baseline**
- Lipids, lymphocytes, neutrophils, hemoglobin, LFTs

RITUXIMAB

bdMARD

Drug	MOA	AEs	D/I
Rituximab (B-cell depletor)	Selectively depletes CD20+ B cells	• Flushing, HTN, hypotension, peripheral edema, angina • Mucocutaneous reactions (e.g. SJS/TEN) • Night sweats • N/D • Chills, fatigue, headache • Hep B virus reactivation • Infection • Infusion reactions (i.e. angioedema, bronchospasm) • <u>Rare</u>: PML	• Immunosuppressants • Tacrolimus, pimecrolimus C/I: hypersensitivity to murine proteins, CHO cell proteins, history of PML, active infection



CLINICAL PEARLS

- Administered IV

BENEFITS

- DMARD of choice for patients with history of lymphoproliferative disorder
- Can be used alone or with csDMARD, but not with other bDMARDs
- Moderate to severe RA with inadequate response to csDMARD and anti-TNF α agents

LIMITATIONS

- Very expensive, adverse reactions

MONITORING

- Baseline TB screening, CBC, LFTs, creatinine, hep B/C serology
- Recommend methylprednisolone (or hydrocortisone), acetaminophen and diphenhydramine administered before each infusion

TOCILIZUMAB

bdMARD

DRUG	MOA	AEs	D/I
TOCILIZUMAB	Anti-IL-6 receptor antibody	- Increased serum cholesterol - Constipation - Injection site reaction - GI perforation - Infection risk - Infusion-related reaction - HTN, headache - Hepatic injury	- Immunosuppressant agents - Tacrolimus, pimecrolimus C/I: Active infection



CLINICAL PEARLS

BENEFITS

- Offers a novel mechanism of action for patients with failure of traditional anti-TNF alpha therapy
- Indicated **for moderate to severe RA**
- Effective as monotherapy without MTX or other DMARDs (can be used with MTX)

LIMITATIONS

- Very expensive
- New or reactivation of infection

MONITORING

- Screen for TB, LFTs, abdominal symptoms

ABATACEPT

bdMARD

DRUG	MOA	AEs	D/I
ABATACEPT	T cell activation inhibitor	• HTN • Hypermagnesemia • Nausea • Headache • AKI • Infection (i.e. TB, sepsis)	• Immunosuppressant agents • Tacrolimus, pimecrolimus • C/I: patients with or at risk of sepsis, active infection



CLINICAL PEARLS

BENEFITS

- May be used as monotherapy or in combination with **DMARD therapy**
- Used in moderate to severe pts that have not responded to DMARDs or an anti-TNF alpha

LIMITATIONS

- May cause false high blood sugar levels due to maltose in IV administration

MONITORING

- Baseline screen for hepatitis and TB

IMMUNIZATIONS & TB SCREENING

DMARDs and glucocorticoids have immunosuppressant effects, which can increase a patient's risk for infections – vaccination records should be reviewed and updated prior to treatment initiation

Live Vaccines

- Avoid in patients taking csDMARDs, tsDMARDs, or bDMARDs
- If indicated, administer at least **4 weeks** prior to initiating DMARDs
- Wait until at least **3 months** after discontinuation of DMARDs to administer live vaccines

Inactivated Vaccines

- Administer at least **2 weeks** prior to initiation of DMARDs to optimize immunogenicity
- If not feasible, inactivated vaccines may be administered during treatment with DMARDs (some studies suggest holding the agent around the time of vaccination while others are contradictory)

Tuberculosis Screening

- All patients must be screened for tuberculosis prior to treatment with a bDMARD or tsDMARD

HIGH-RISK CONDITIONS/SPECIFIC PATIENT POPULATIONS

Congestive heart failure: In patients classified with NYHA III or IV CHF with inadequate response to csDMARDs, it is conditionally recommended to add **non-TNF alpha inhibitor bDMARD or tsDMARD** over TNF alpha inhibitor

Pulmonary disease: **methotrexate** is conditionally recommended versus other DMARDs for patients with clinically diagnosed airway disease with moderate-high disease activity

Previous serious infection*: addition of **csDMARDs** is conditionally recommended over addition of a bDMARD or tsDMARD for patients with a serious infection within the previous 12 months who have moderate-high disease activity despite csDMARD monotherapy

*Serious infection refers to an infection requiring hospitalization or antibiotic treatment

SPECIFIC PATIENT POPULATIONS

Pregnancy

- **Methotrexate** → **CONTRAINDICATED**
 - R/O pregnancy prior to starting treatment, use reliable contraception
 - Wait at least 3 months following D/C before attempting conception and supplement with folic acid
- **Leflunomide** → **CONTRAINDICATED**
 - R/O pregnancy prior to starting treatment
 - Use reliable contraception before, during treatment and for up to 2 years after D/C
 - If unable to wait 2 years – perform washout with cholestyramine or activated charcoal (must wait until plasma levels are below 0.02 mg/L for minimal risk of teratogenicity)
- **Sulfasalazine** → considered safe, supplement with folic acid 1 mg daily
- **Hydroxychloroquine** → considered safe
- **Certolizumab** → compatible with all trimesters
- **Glucocorticoids** → considered safe, use lowest dose possible
- **Adalimumab, infliximab, etanercept**
→ can be used but generally should be discontinued in 3rd trimester
- **Rituximab, tocilizumab** → may be used with caution only if needed
- **NSAIDS** → C/I in third trimester
- **JAK Inhibitors** → C/I due to insufficient data

Breastfeeding

- Glucocorticoids, NSAIDs, hydroxychloroquine, sulfasalazine → **compatible** (do not use sulfasalazine if infant has hyperbilirubinemia)
- Methotrexate, leflunomide → **CONTRAINDICATED**
- bDMARDs → **compatible**
- tsDMARDs → limited data

TAPERING/DISCONTINUATION

1. Patients at target or remission for at least 6 months:

- Consider stepwise reduction in the dose of b/tsDMARD (without complete discontinuation), provided patients can rapidly access rheumatology care and re-establish their medications in case of a flare
- **Conditional recommendation:** avoid tapering for patients where rapid access to care is challenging
- If patient is on both csDMARD and b/ts DMARD, either one can be tapered first, with shared decision making **April 2026 CRA update**

2. **Conditional recommendation:** Gradual discontinuation of sulfasalazine preferred over gradual discontinuation of hydroxychloroquine for patients who are taking triple therapy

MONITORING

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Signs and symptoms (e.g. swelling, joint stiffness, fatigue)	Reduce/eliminate	1 week – 1 month
Range of Motion, function	Improve/normalize	1-2 months
Pain	Reduce/eliminate	1 week (with adjunct therapy) to 2 months (significant reduction in pain)

Safety Endpoints		
Parameter	Degree of Change	Timeframe
Adverse effects from drug (e.g. diarrhea, upset stomach) and worsening symptoms	Prevent/minimize	Duration of therapy
Lab tests (e.g. CBC, LFTs, SCr; these may be drug specific)	Minimal/none (<i>Baseline blood work should be WNL</i>)	Duration of therapy



Laboratory work required prior to initiation of drug therapy

TAKEAWAY

- RA is characterized by morning stiffness lasting >1 hour and has a symmetrical presentation; symptoms improve with movement
- PCT for RA is determined based on disease activity
 - For LOW disease activity: HCQ is preferred over SSZ
 - For HIGH disease activity OR presence of poor prognostic factors: MTX is preferred. Maximize MTX before switching therapies. If there is no improvement in 3-6 months, add-on bDMARD
 - Poor prognostic factors = presence of autoantibodies, high disease activity, early erosions or failure of two csDMARDs
- Since the onset of action of DMARD's can take weeks, adjunctive NSAIDs are indicated for pain relief for the first 3 months of treatment
- Before starting DMARD therapy, baseline blood work and liver/kidney function tests should be done and repeated periodically throughout treatment

CASE 1

RA is a 75-year-old female you see regularly in the family medicine clinic. Today, she looks more tired than usual. By speaking with RA, you find out that she is being worked up for the pain she has been experiencing. The test results are not back yet. RA has a past medical history of dyslipidemia, diabetes mellitus, and hypertension. She is currently taking atorvastatin 10mg PO qhs, metformin 500 mg PO BID, gliclazide MR 30 mg PO daily, hydrochlorothiazide 12.5 mg PO daily, ramipril 2.5 mg PO daily, ASA 81 mg PO daily and multivitamin for ages 50+ PO daily. Her blood pressure is 128/80 mmHg, HR: 68 bpm, and A1C 7.3%. For the last 6 months she has felt exhausted. She has morning stiffness (lasts longer than an hour), especially in her hands. She is allergic to penicillin.

1. Which of the following interventions would be the most appropriate next-step for Mrs. RA?

- a) Start a biologic
- b) Start a csDMARD
- c) Perform baseline CBC, liver and renal biochemistry, RF, anti-CCP, ESR, CRP, and radiographs of hand/feet
- d) Start triple therapy with MTX, anti-TNF, and cyclosporine

"I'm frustrated. It takes a long time to get dressed and I find knitting very difficult."



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2. What would be a reasonable timeframe to assess disease activity after starting first-line therapy for Mrs. RA?
- a) Every 1-2 weeks
 - b) Every 2-3 weeks
 - c) Every 1-3 months
 - d) Every 1-2 years



CASE 1

RA is a 75-year-old female you see regularly in the family medicine clinic. Today, she looks more tired than usual. By speaking with RA, you find out that she is being worked up for the pain she has been experiencing. The test results are not back yet. RA has a past medical history of dyslipidemia, diabetes mellitus, and hypertension. She is currently taking atorvastatin 10mg PO qhs, metformin 500 mg PO BID, gliclazide MR 30 mg PO daily, hydrochlorothiazide 12.5 mg PO daily, ramipril 2.5 mg PO daily, ASA 81 mg PO daily and multivitamin for ages 50+ PO daily. Her blood pressure is 128/80 mmHg, HR: 68 bpm, and A1C 7.3%. For the last 6 months she has felt exhausted. She has morning stiffness (lasts longer than an hour), especially in her hands. She is allergic to penicillin.

3. Once target therapy is reached with methotrexate and Mrs. RA is in remission for at least 6 months, which of the following can be considered?
- a) Discontinue methotrexate and reassess her symptoms in 1 month
 - b) Reduce the dose of methotrexate gradually while continuously reassessing her symptoms
 - c) Switch methotrexate to sulfasalazine
 - d) Switch methotrexate to hydroxychloroquine



CASE 2

KP is a 65-year-old male who has been on prednisone 50mg for the last 2 months for a rheumatoid arthritis flare. His doctor wants you to help taper down and then stop his prednisone dose as the flare has subsided and Mr. KP is stable on leflunomide. He would like you to reduce his dose by 5 mg every 2 weeks for the first 6 weeks, then by 5 mg every week until he gets to 10 mg once daily. Then he would like you to taper his dose by 2.5mg every week until the end of the taper. You only carry 5mg tablets of prednisone.

How many tablets of prednisone will Mr KP need to complete his taper?

- a) 329
- b) 497
- c) 1097
- d) 507



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CHANGE LOG

June 2025

- Slide 32: Changed folic acid recommendation with SSZ from 5mg to 1mg as per SOGC

August 2025

- Formatting changes made throughout
- Removed mention of azathioprine as no longer in guidelines
- Slide 22: Modified to reflect all JAK inhibitors approved in Canada
- Slide 27: Added the definition of 'serious infection'
- Slide 33: Removed "New" from 2023 update
- Slide 36: Minor wording changes

September 2025

- Slide 8: Removed 'FYI'

November 2025

- Minor grammar, phrasing and formatting changes throughout
- Slide 3: Updated legend
- Slide 10: Replaced 'hyperlipidemia' with 'dyslipidemia'
- Slide 13: Added leflunomide to low disease activity algorithm. Added note on trialling SC MTX if PO does not achieve target
- Slide 14: Minor update to slide
- Slide 27: Removed numbers, unnecessary
- Slide 31: Updated takeaway
- Slide 34: Minor update to answer choices

April 2026

- Slide 13: Added 'DMARD Naïve patients' to title
- Slide 22: added clinical pearl about HZ vaccination
- Slide 28: Updated/simplified recommendations in pregnancy
- Slide 29: Added a statement as per CRA 2026 update and simplified recommendations
- Slide 36: Updated references
- Minor formatting and phrasing changes made throughout